

RetenNet Feature Intelligence

Strategic Feature Explanations & Business Logic

This document details the critical engineered features used in the RetenNet Churn Prediction model. Standard transactional metrics often fail to capture the nuances of customer behavior, leading to high false-positive rates. By injecting mathematical boundaries and business logic into the feature space, the model achieves a high signal-to-noise ratio and explicitly distinguishes between active dormancy and true churn.

1. Mitigating Mathematical Distortion

The Recency Cap (days_since_last_txn)

The Problem: Missing recency values were historically imputed with **9999**. When passed through a Min-Max Scaler, this extreme outlier compresses all normal transaction recencies (e.g., 5 days vs. 90 days) into a microscopic mathematical space, destroying the geometric variance required by the Support Vector Machine (SVM) RBF kernel.

The Solution: We cap the maximum recency at 180 days (the maximum extent of our sliding window analysis). This explicitly restores proportional distance in the feature space.

```
days_since_last_txn = min(days_since_last_txn, 180)
```

2. Boolean Indicator Functions (Interaction Features)

To explicitly guide the SVM through non-linear behavioral intersections, we defined hard-coded binary flags based on the intersection of 30-day transaction frequencies (T_{30}) and average 30-day balances (B_{30}).

The Passive Saver (is_passive_saver)

Logic: `txn_count_30d == 0 AND avg_daily_balance_30d > 10`

Business Context: Identifies users who have no recent transactional activity but retain significant capital in their accounts. Standard models falsely flag these users as churners. This explicit feature protects "active dormancy" and prevents the company from wasting retention budgets on high-balance savers.

The True Abandoner (is_true_abandoner)

Logic: `txn_count_30d == 0 AND avg_daily_balance_30d ≤ 1`

Business Context: Identifies true churn risk. These are users who have completely stopped transacting and have drained all liquid capital from their accounts. By pre-splitting this coordinate space, the SVM no longer has to implicitly learn the relationship between zero transactions and zero balances.

The New Onboarding (is_new_onboarding)

Logic: `txn_count_30d == 0 AND account_tenure_days ≤ 14`

Business Context: New users often have a delay between account creation and their first transaction. This flag explicitly tells the model to excuse the lack of T_{30} activity if the user's overall tenure is less than two weeks, preventing false-positive flags during the onboarding phase.

3. Sliding Window Velocity Metrics

Balance Trend Ratio (`balance_trend_ratio`)

Concept: Calculates the ratio of the 30-day average daily balance to the 90-day average daily balance.

Business Context: Captures the *velocity* of capital movement. A ratio of < 1.0 indicates a customer is actively draining their account over time, signaling a slow-fade churn pattern long before their balance hits absolute zero.